

EconomicBridge — Platform Briefing for the Director-General, NASRDA

AI & Satellite Intelligence for Agriculture, Food Security & Aid Delivery Operated by Bizra Farms Integrated Nigeria Ltd

1. What it is — in one line

A multi-tenant satellite-intelligence platform covering **52 West African administrative units** (36 Nigerian states + FCT + 15 ECOWAS countries; 10 live pilots today), turning **live NASA + Copernicus satellite feeds** into actionable, **per-LGA** intelligence across seven modules — with a full **provenance trail** on every layer.

Pilots live now: Kebbi · Benue · Plateau · Kaduna · Niger · Zamfara · Nasarawa · FCT · Ghana · Senegal.

2. The core differentiator — multi-sensor satellite fusion

No single satellite tells the whole story. EconomicBridge **fuses four complementary sensors**, each covering the others' blind spots:

Sensor	What it sees	Covers the gap of
Sentinel-2 (optical, NDVI)	vegetation health / loss	(blind at night + under cloud)
Sentinel-1 (radar, SAR)	land-surface change, all-weather, day & night	optical's cloud/night blindness
NASA FIRMS (thermal)	active fires / burning	slow-onset change
VIIRS Black Marble (night-lights)	new human activity, every night, year-round	seasonal blind spots of the above

Satellite constellation — revisit, resolution & role

Satellite / product	Sensor	Resolution	Revisit	Swath	Data latency	Provider
Sentinel-2 (MSI)	Optical, 13 bands	10 m (NDVI bands)	~5 days	290 km	hours – 2 days	ESA / Copernicus
Sentinel-1 (SAR)	C-band radar	~10 m (IW GRD)	~6 days	250 km	hours – 1 day	ESA / Copernicus
NASA FIRMS	VIIRS / MODIS thermal	375 m (VIIRS)	multiple / day	—	~3 h (near-real-time)	NASA LANCE/EOSDIS
VIIRS Black Marble (VNP46A2)	Day/Night Band	~500 m	nightly (daily)	—	~1 day – 1 week	NASA / LAADS DAAC
WorldPop	Modelled population grid	~100 m	annual	—	—	WorldPop (CC-BY)

Notes: Sentinel revisit reflects the two-satellite constellation (S2A/S2B; S1A/S1C). SAR is all-weather, day & night (cloud-penetrating); the optical NDVI is masked per-pixel for cloud. FIRMS and Black Marble are the high-cadence layers that keep the platform alert between Sentinel passes. Higher-resolution imagery (e.g. NASRDA **NigeriaSat / NCRS**) would sharpen the per-LGA indicators to sub-field scale on top of this open base.

How the fusion works (per location, not per state):

1. **Per-LGA evaluation** — every Local Government Area is assessed from *its own* satellite time-series, not a single state-wide average that washes out local events. 2. **Revisit-matched refresh** — each LGA is re-scanned on the satellite's natural ~6-day revisit, so coverage is always fresh without wasting compute. 3. **Corroboration scoring** — a *single* signal raises a cautious watch; *corroborating* signals (e.g. vegetation loss **and** radar disturbance **and** a new night-light) escalate confidence to high/critical. This is what separates a real event from sensor noise. 4. **Honest provenance** — every output is tagged with its sensors, acquisition source, and a *model-derived vs measured* flag. Nothing is fabricated.

> The result: a system that stays alert **year-round** — radar works through the wet season, night-lights flag new activity in any season, and fire/optical add corroboration when present.

3. The seven modules — how each works

Module 03 — Farmland Protection (flagship)

- **Detects:** encroachment & land-surface disturbance (grazing incursion, clearing, new camps/mining) as a 24–72h conflict-risk indicator.
- **Fuses:** Sentinel-1 SAR + Sentinel-2 NDVI + NASA FIRMS + **VIIRS new-light**.
- **New amendment:** rebuilt from one state-wide alert to **full per-LGA coverage across all 10 states**, refreshing daily; a **year-round "new light in dark farmland"** signal now flags fresh human activity even when vegetation and radar are quiet.

Module 04 — CropGuard

- **Detects:** crop/vegetation health (satellite) + crop disease (AI).
- **Fuses:** Sentinel-2 NDVI + Sentinel-1 SAR + **a trained ResNet-50** that diagnoses disease from a single leaf photo (87% validation accuracy).
- **New amendment:** **statewide per-LGA crop health** — every LGA now carries a live Sentinel-2 NDVI health reading (healthy → moderate → stressed → poor), so CropGuard covers the whole state, not only photo-sampled points. Plus field-record keeping (every Farm Check + leaf diagnosis saved and recallable by state/LGA/crop).
- **The honest split:** satellite tells you **where** crops are stressed (per-LGA); a leaf photo tells you **which** disease (point-level) — a physical limit of remote sensing, and a credible one.

Module 05 — ShockGuard

- **Detects:** floods (sharp radar drop = standing water) and droughts (sharp NDVI drop).
- **Fuses:** Sentinel-1 SAR + Sentinel-2 NDVI, **per-LGA**.
- **New amendment:** rebuilt to **per-LGA scanning of every LGA**; correctly shows **"all-clear, monitoring live"** when there are no shocks (it is early wet season — vegetation greening, no flood signatures), and lights up the affected LGAs the moment a real shock appears. Historical examples are clearly labelled.
- **Coverage:** all three satellite modules now reach every LGA — Farmland (per-LGA encroachment), CropGuard (per-LGA health), ShockGuard (per-LGA monitoring).

Module 01 — Economic Visibility (Poverty mapping)

- **Detects:** under-electrified / under-served settlements.
- **Fuses:** **VIIRS Black Marble** night-light radiance + WorldPop population.
- **New amendment:** now reads **true per-pixel night-light radiance** from NASA granules (validated live — capitals bright, rural dark), replacing an earlier modelled proxy.

Module 06 — Economic Mobility · Module 07 — SkillsBridge · Module 02 — Aid Coordination

- **Mobility:** World Bank income/employment indicators + Nigeria NLSS calibration.
 - **Skills:** UNICEF GIGA real school access & connectivity.
 - **Aid:** OCHA HDX humanitarian operational-presence coverage.
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4. Provenance & licensing — "is it real, and can we commercialise it?"

Yes on both. The platform carries an open, in-product **Data Sources & Provenance** screen showing, per layer: the satellite + product, provider, licence, the exact attribution, live-vs-modelled status, and refresh cadence.

Source	Commercial use	Obligation
Copernicus / Sentinel-1 & 2 (ESA/EU)	Allowed (free, full & open policy)	Attribution: "Contains modified Copernicus Sentinel data [year]"

NASA — FIRMS, VIIRS Black Marble, MODIS	Allowed (US public domain)	Cite source (e.g. <i>NASA Black Marble VNP46A2, LAADS DAAC</i>)
WorldPop / World Bank	Allowed (CC-BY 4.0)	Attribution

> **Compute:** live Sentinel feeds draw on the Copernicus free "General" allowance (30,000 Processing Units/month); per-LGA coverage is revisit-matched to stay well inside it.

5. Why it is credible to a satellite agency

- **Genuine data only** — every satellite layer is read live from public NASA + Copernicus sources. No simulated or placeholder values on the map.
 - **Per-LGA, not per-state** — local resolution, real coordinates, real place names.
 - **Honest when quiet** — a disaster module that shows "all-clear" when there are no disasters is *more* credible than one that always shows alerts.
 - **Auditable** — the provenance screen is the single place that answers "where is this from?"
 - **Production-hardened** — automatic rate-limit handling, daily self-refreshing feeds, and a deploy + migration pipeline already proven on AWS.
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6. Where NASRDA fits

- **Higher-resolution imagery (NigeriaSat / NCRS):** would sharpen our ROI/LGA indicators to **sub-field** resolution — the natural next layer on top of the open Sentinel/VIIRS base.
 - **ASEICC demonstration platform:** ready to showcase the live system.
 - **Joint validation:** ground-truth campaigns to calibrate the encroachment + crop-stress models against field reality.
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